DOST Form 5

A – PROJECT WORKPLAN

(1) Program Title: N/A
(2) Project Title: Establishing RadioJOVE and SkyNet for Astronomical and Exoplanet Research: Application of White Noise Analysis of Curated Datasets
(3) Project Duration (number of months): 12 **(4) Project Start Date:** September 2025 **(5) Project End Date:** August 2026

(6) OBJECTIVES	(7) TARGET ACTIVITIES	(8) TARGET ACCOMPLISHMENTS (quantify, if possible)	Y1 (2025)	Y2 (2026)			Total
			Q4	Q1	Q2	Q3	
1. To establish and operate the first RadioJOVE-based radio observation station in the Philippines. 2. To develop and validate a data analysis workflow that processes SkyNet telescope data alongside NASA's Exoplanet Archive and Harvard's DIY planet search. .	Mobilization of project staff	(1) Local Recorded Data from RadioJOVE (2) Local recorded data from submissions in SkyNet	x				
	Procurement of equipments and assembly		x				
	Pipeline Process development		x	x			
	Data Collection using RadioJOVE and SkyNet				x	x	
	Training in RadioJOVE and SkyNet systems*		x	x			
	Consultations with experts in related astronomical fields		x	x	x	x	
	Capacity building for local science and research community that highly encourage the participation of both women and men in all activities					x	
3. To generate, organize, and analyze datasets related to RadioJove radio astronomy and exoplanet observations and apply stochastic white noise analysis for scientific pattern detection.	Curate astronomical datasets	(1) Curated Astronomical Datasets from RadioJOVE and SkyNet (2) At least 1 scientific paper on stochastic analysis of astronomical datasets (3) Atleast 10 LAS for project-related topics		x	x	x	
	Conduct of Jagna Space Encounters 2.0 workshop on techniques in analysis of astronomical datasets.			x			
	Apply stochastic white noise analysis on applicable datasets				x	x	
	Consultations with experts on stochastic white noise analysis		x	x	x	x	
	Writing of Learning Activitiy Sheets (LAS) to project-related topics*				x	x	

*This will ensure the establishment of protocols for using RadioJOVE and SkyNet. After this project, interested senior high school students can have a hands-on experience on their Research courses. Researchers and student-researchers can also continue to collect and process data even after the project has concluded.

B – EXPECTED OUTPUTS

(1) Program Title: N/A

(2) Project Title: Establishing RadioJOVE and SkyNet for Astronomical and Exoplanet Research: Application of White Noise Analysis of Curated Datasets

(3) Project Duration (number of months): 12 **(4) Project Start Date:** September 2025 **(5) Project End Date:** August 2026

(9) EXPECTED OUTPUTS (6Ps)	Objectively Verifiable Indicators (OVIs)				
	Y1 (2025)	Y2 (2026)			Total
	(Q1) Sept 2025 – Nov 2025	(Q1) Dec 2025 – Feb 2026	(Q2) Mar 2026 – May 2026	(Q3) Jun 2026 – Aug 2026	
Publications				Curated astronomical datasets	• Curated astronomical datasets
Patents/IP					
Products					
People Services	3 science researchers	3 science researchers	3 science researchers	3 science researchers	• 3 science researchers
Places and Partnerships	Consultants from different institutions	Consultants from different institutions	Establihed Contact at NASA RadioJOVE Project Team and SkyNet UNC	Establihed Contact at NASA RadioJOVE Project Team and SkyNet UNC	• Consultants from different institutions • Establihed Contact at NASA RadioJOVE Project Team and SkyNet UNC
Policy					
(10) POTENTIAL IMPACTS (2Is)					
Social Impact	• Building linkages with astronomy communities and institutions. • Enhancing local capacity in astrophysical data handling.	• Building linkages with astronomy communities and institutions. • Enhancing local capacity in astrophysical data handling.	• Building linkages with astronomy communities and institutions. • Enhancing local capacity in astrophysical data handling.	• Building linkages with astronomy communities and institutions. • Enhancing local capacity in astrophysical data handling.	• Building linkages with astronomy communities and institutions. • Enhancing local capacity in astrophysical data handling.
Economic Impact				• Creating opportunities for local suppliers and technical staff through radio-telescope assembly and maintenance. • Developing workforce skills in data science that can be applied in national industries beyond astronomy.	• Creating opportunities for local suppliers and technical staff through radio-telescope assembly and maintenance. • Developing workforce skills in data science that can be applied in national industries beyond astronomy.

C – RISKS AND ASSUMPTIONS

(1) Program Title: N/A
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OBJECTIVES	(11) RISKS AND ASSUMPTIONS	(12) ACTION PLAN (use separate sheet if necessary)
1. To establish and operate the first RadioJOVE-based radio observation station in the Philippines.	**Risks:** Equipment delays, safety issues in field setup. **Assumptions:** Materials and kits are available locally or via import; safe and secure site access is ensured.	Pre-order critical components, implement field safety protocols, train staff on setup and calibration.
2. To develop and validate a data analysis workflow that processes SkyNet telescope data alongside NASA's Exoplanet Archive and Harvard's DIY planet search.	**Risks:** Difficulty in integrating datasets (SkyNet, NASA, DIY tools), software compatibility issues, limited technical expertise. **Assumptions:** Access to SkyNet, NASA Exoplanet Archive, and DIY tools remains stable; collaboration support from partners continues.	Assign a dedicated data pipeline team,, schedule regular testing, and consult with astronomy data specialists for troubleshooting.
3. To generate, organize, and analyze datasets related to RadioJove radio astronomy and exoplanet observations and apply stochastic white noise analysis for scientific pattern detection.	**Risks:** Incomplete or noisy datasets, errors in stochastic modeling, limited expertise in advanced statistical methods. **Assumptions:** Access to RadioJOVE, SkyNet, and NASA datasets is consistent; experts are available for consultation.	Establish data curation protocols, run pilot analyses before scaling, and conduct consultations on stochastic methods.

DOST Form 5
PROJECT WORKPLAN, EXPECTED OUTPUTS, RISKS AND ASSUMPTIONS

I. General Instruction: Submit through the DOST Project Management Information System (DPMIS), http://dpmis.dost.gov.ph, the project workplan, expected outputs, and risks and assumptions together with the project proposal. Also, submit four (4) copies of these forms together with the proposal. Use Arial font, 11 font size.

II. Operational Definition of Terms:

1-2. Program/Project Title- the identification of the Program and its component projects.

3-5. Project Duration and **Project Start/End Date**- refer to the grant period or timeframe that covers the approved start and completion dates of the project, and the number of months the project will be implemented.

6. Objectives- statements of the general and specific purposes to address the problem areas of the project.

7. Target Activities- set of fixed works that needs to be conducted for the achievement of the project objectives.

8. Target Accomplishments- measurable and positive results of completing project activities.

9. Expected Outputs- deliverables of the project based on the 6Ps metrics (Publications, Patents/Intellectual Property, Products, People Services, Places and Partnerships, and Policy).

a. **Publication**- published aspect of the research, or the whole of it, in a scientific journal for peer review. (get Definition from DOST OUTcomes)

b. **Patent/Intellectual Property-** proprietary invention or scientific process for potential future profit.

c. **Product-** invention with a potential for commercialization.

d. **People Services-** people or groups of people, who receive technical knowledge and training.

e. **Partnership-** linkage forged because of the study.

f. **Policy-** science-based policy crafted and adopted by the government or academe as a result of the study.

10. Potential Impacts

a. Social Impact- refers to the effect or influence of the project to the reinforcement of social ties and building of local communities.

b. Economic Impact- refers to the effect or influence of the project to the commercialization of its products and services, improvement of the competitiveness of the private sector, and local, regional, and national economic development.

11. Risk- refers to an uncertain event or condition that its occurrence has a negative effect on the project.

Assumption- refers to an event or circumstance that its occurrence will lead to the success of the project.

12. Action Plan- proposed activities to address the risks and assumptions